

12 - Logistic Regression

Artificial Intelligence Lab Manual - Clean Study Edition

Learning Objectives

- Model binary class probability with the sigmoid function.
- Train a logistic-regression classifier and interpret its decision boundary.
- Evaluate predictions with confusion-matrix-based metrics.

Background

Logistic regression is a linear probabilistic classifier. It estimates a class probability and then applies a threshold. Despite its name, it is used for classification rather than ordinary continuous-value regression.

Core Concepts

Concept	Meaning in this lab
Logit	Linear score $z = w^T x + b$.
Sigmoid	Maps a real-valued score into the interval (0,1).
Cross-entropy	Common loss for probabilistic binary classification.
Decision threshold	Probability cutoff used to convert probabilities to labels.
Regularization	Penalty that can limit excessive coefficient magnitude.

Guided Procedure

1. Prepare a binary classification dataset and standardize numeric features.
2. Compute sigmoid probabilities from a linear score.
3. Train with gradient descent from small initial weights or use scikit-learn after implementing the core equations.
4. Evaluate accuracy, precision, recall, and confusion matrix.
5. Change the decision threshold and observe the precision-recall trade-off.

Reference Implementation

Use the following as a starting point. Read and modify it rather than treating it as a final answer.

```
import numpy as np

def sigmoid(z):
    return 1.0 / (1.0 + np.exp(-z))

def train_logistic(X, y, lr=0.05, steps=1000):
    w = np.zeros(X.shape[1])
    b = 0.0
    for _ in range(steps):
        p = sigmoid(X @ w + b)
        w -= lr * (X.T @ (p - y)) / len(y)
        b -= lr * (p - y).mean()
    return w, b
```

Lab Exercises

Task 1. Train the from-scratch model and compare with `sklearn.linear_model.LogisticRegression`.

Task 2. Plot a decision boundary for a two-feature dataset.

Task 3. Evaluate thresholds 0.3, 0.5, and 0.7.

Task 4. Explain why data leakage can occur if scaling is fit on the full dataset before splitting.

Suggested Deliverables

- Notebook or Python file containing the completed implementation.
- Short result table or output evidence for each required comparison.
- A concise interpretation of what changed and why; do not submit output without explanation.

Review Questions

1. Why does logistic regression use a sigmoid?
2. What does the decision threshold control?
3. When can a linear decision boundary be too restrictive?

Completion check: You should be able to explain the algorithm or workflow in your own words, reproduce the main result, and identify at least one limitation.